Praison John Raj S

Full Stack Developer | Real-Time Communications & Cloud

Chennai, India · +91 84282 73545 · praisonjohnraj.s@gmail.com

praisonjohnrajs.github.io/portfolio · linkedin.com/in/praison-john-raj-12dec2001 · github.com/PraisonSWDP

PROFESSIONAL SUMMARY

Full Stack Developer with 3.5+ years of experience designing and shipping enterprise contact-center software end to end. Specialized in real-time voice and messaging systems (WebRTC, SIP.js, Asterisk, FreeSWITCH), Python/FastAPI and Node.js microservices, and containerized cloud infrastructure on AWS and Kubernetes. Proven ownership from architecture and security design through CI/CD and production operations.

CORE SKILLS

Languages	Python, JavaScript (ES6+), PHP, SQL, Lua
Frontend	React.js, Vite, Tailwind CSS, SIP.js, WebRTC, Socket.IO, HTML5, CSS3
Backend	FastAPI, Node.js, Express.js, SQLAlchemy, Pydantic, REST APIs, WebSockets
Telephony	Asterisk (AMI), FreeSWITCH (ESL), SIP / PJSIP, IVR, call routing, live monitoring
Data & Messaging	MySQL, MongoDB, Redis (Pub/Sub, caching), Apache Kafka
Cloud & DevOps	AWS (EC2, S3, Lambda, Route 53, API Gateway), Docker, Kubernetes, GitHub Actions CI/CD, Nginx, Apache2, Linux
Security	JWT, RBAC, 2FA/MFA (TOTP), bcrypt password hashing
Integrations	Salesforce CTI, Freshworks, WhatsApp Business messaging, AI voice (ElevenLabs)

PROFESSIONAL EXPERIENCE

Software Developer — Pulse Telesystems Private Limited

Apr 2023 – Present

Chennai, India

- Lead full-stack development of **ConnectHub**, an enterprise cloud contact-center platform covering inbound, outbound and blended campaigns, CRM, IVR, live monitoring and reporting.
- Architected a **microservices backend** of 25+ Python/FastAPI and Node.js services (auth, call routing, billing, reporting, IVR blast, chat), containerized with Docker and deployed on **Kubernetes** with GitHub Actions promotion pipelines.
- Built a browser-based **WebRTC softphone** with SIP.js, integrated with Asterisk and FreeSWITCH, removing the need for desktop softphone clients.
- Engineered real-time supervisor dashboards (agent states, queues, listen/whisper/barge) using WebSockets, Socket.IO, Redis Pub/Sub, Asterisk AMI and FreeSWITCH ESL.
- Designed event-driven processing with **Apache Kafka** producers/consumers for call events, conference handling, SMS/WhatsApp delivery reports and billing.
- Delivered CRM telephony integrations including **Salesforce CTI click-to-call** and Freshworks, plus WhatsApp messaging and AI-driven voice agents.
- Implemented JWT authentication, RBAC and TOTP-based 2FA/MFA; deployed and operated production infrastructure on AWS (EC2, S3, Lambda, Route 53, API Gateway) behind Nginx/Apache2.

KEY PROJECTS

Emergency IVR & Multichannel Alert System (sole developer)

2026

FastAPI, Kafka, FreeSWITCH ESL, Lua, AWS S3, Docker, Kubernetes, GitHub Actions

- Designed and built end to end a high-volume emergency outbound calling system: an event-driven Kafka consumer originates calls through FreeSWITCH ESL across multiple carrier gateways.
- Developed multilingual IVR flows in Lua with DTMF response capture and on-demand text-to-speech, plus SMS and WhatsApp alerts with delivery-report tracking.
- Automated call recording with upload to AWS S3, call status APIs and exportable reports; resolved ESL concurrency deadlocks and thread-safety crashes for stable production use.

Switch Automation System — Network Device Management

Node.js, React.js, MongoDB, WebSockets

- Centralized administration of network switches with real-time status tracking, automated configuration backups, task scheduling and batch configuration.

EDUCATION

Master of Computer Applications (MCA) — **Anna University, Tamil Nadu** 2023 – 2024

Bachelor of Computer Science (BCS) — **Madurai Kamaraj University, Tamil Nadu** 2019 – 2022